Pedro Henrique Souza Balbino | Senior Software Engineer

Recife, PE – Brazil (Remote)

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5+ years | BRT/UTC-3 – Full US East Coast overlap

Summary

Senior AI Developer and Backend Engineer building scalable distributed systems and AI-powered solutions in **Go**, **TypeScript**, and **Python**. Leading a multi-agent AI system at Pearson Education processing **699K+ records** with **340% ROI** and **$80K annual savings**. Expertise in Amazon Web Services (AWS), Kubernetes, Terraform, and production RAG/LLM pipelines. **AWS Certified Solutions Architect – Professional**.

Key Impact

699K+	340% ROI	95%+
student records processed	$80K annual savings	OCR extraction accuracy
99.1% uptime	multi-agent RAG pipeline	200+ page documents

Experience

Stefanini LATAM | Pearson Education — **Remote**

Senior AI Developer — *Jul 2025 – Present*

- Architected **4-agent AI system** (Triage, Franchise, Analytics, Response) using LangGraph StateGraph, processing 699K+ student enrollment records with 340% ROI and $80K annual savings
- Built bilingual WhatsApp chatbot with Azure OpenAI GPT-4: **99.2% query success**, 2.8s avg response
- Improved AI knowledge base accuracy by 40%; reduced manual documentation by 60%
- *Stack: AWS Bedrock, LangChain, LangGraph, Azure OpenAI, Python, FastAPI, SQL Server, Redis, K8s*

Globalsys | Grupo AR&CO (Arezzo + Soma) — **Remote**

Software Engineer — *Nov 2022 – Oct 2025*

- Designed scalable microservices for Brazil's largest fashion retail group handling high-volume transactions
- Implemented event-driven architecture with Apache Kafka, AWS SQS/SNS, multi-provider CRM integrations
- *Stack: TypeScript, NestJS, Node.js, Go, GraphQL, Apache Kafka, SQL Server*

Ultracon | GlobalHitss (América Móvil/Claro) — **Remote**

Software Engineer — *Oct 2024 – Sep 2025*

- Architected document processing with AWS Bedrock for Latin America's largest telecom
- Automated legal workflows with NLP/LLM integration using serverless Go and Python
- *Stack: Go, Python, AWS Lambda, Step Functions, Terraform*

aca.so — **Remote**

Software Engineer — *Jun – Nov 2023*

Graph-based social platform with Amazon Neptune/Gremlin; async FastAPI APIs; junior mentorship. *Stack: Python, FastAPI, Go, Neptune, ElasticSearch*

RedFox Digital Solutions — **Remote**

Software Engineer — *Jul 2021 – Jun 2023*

Full stack web & mobile – React, Angular, Node.js, Express, PostgreSQL (2 years)

CITi UFPE — **Remote**

Software Analyst — *Jul 2021 – Jun 2022*

React, React Native, Node.js; technical research and junior mentorship

Independent Consulting – BALBINO DEV

ASA Investments (via Sciensa) — **Remote**

Backend Engineer — *Feb 2026 – Present*

- Built Python/FastAPI OCR pipeline processing **200+ page** IRPF tax declarations with **95%+ accuracy**
- Resolved 6 critical extraction bugs – all PRs merged and validated by QA on 10-person team
- *Stack: Python, FastAPI, Azure DevOps, PDFPlumber, Document AI, OCR*

DeliCasa – Smart Vending Machine
Lead Developer, Paused *Jan 2024 – Jun 2025*
Computer vision + IoT: Raspberry Pi, ESP32, VPN security, React Native mobile app

Certifications

2023: **AWS Certified Solutions Architect – Professional** Amazon Web Services
2024: **Google Data Analytics Professional Certificate** Google *(P2E37X4CD5PS)*

Education

UFPE – Centro de Informática **Recife**
BSc Information Systems, 7th semester, 54% *2020 – Present*
Thesis: Observable and self-adaptive architecture for computer vision pipelines in Edge IoT

Technical Skills

Languages: Python (Expert) · Go/Golang (Advanced) · TypeScript (Expert) · JavaScript · SQL
AI/ML: AWS Bedrock · LangChain · LangGraph · RAG · LLM · Azure OpenAI · NLP · MLOps · Document AI
Cloud: Amazon Web Services (AWS) · Terraform (IaC) · Docker · Kubernetes (K8s) · Azure DevOps
Backend: FastAPI · NestJS · Microservices · REST APIs · GraphQL · gRPC · Event-Driven Architecture
Data: PostgreSQL · SQL Server · MongoDB · Neptune · ElasticSearch · Redis · PGVector
Methods: Distributed Systems · Agile/Scrum · CI/CD · Observability · Clean Architecture

Languages

Portuguese: Native
English: Professional *Daily collaboration with international teams*